

C# Developer Practical Exercise: Call Rating Service

Overview

This exercise is designed to assess your backend development skills, including:

- API and service design
- data modelling
- business logic implementation
- code structure and maintainability
- testing
- handling real-world edge cases

No prior telecoms experience is required.

The domain has intentionally been simplified to focus on engineering quality rather than industry knowledge.

Scenario

A company provides telephone services to customers.

Every phone call generates a Call Detail Record (CDR), which contains information such as:

- the customer
- the destination number
- the call duration

Each customer has a rate card defining how much different call types cost.

Your task is to build a service that:

1. accepts call records
 2. applies the correct customer pricing
 3. calculates line-level call costs
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Technical Requirements

Please use:

- C#
- .NET (latest stable version preferred)
- ASP.NET Core
- Entity Framework Core

You may use either:

- SQL Server
- SQLite

You may use additional open-source libraries where appropriate.

Core Concepts

Call Detail Record (CDR)

A CDR represents a completed phone call.

Example:

```
{
  "customerId": 1001,
  "callDate": "2026-05-31T10:15:00Z",
  "destinationNumber": "+441234567890",
  "durationSeconds": 125
}
```

Customer Rate Card

Each customer has their own pricing.

Example:

```
{
  "customerId": 1001,
  "rates": [
    {
      "callType": "UK",
      "costPerMinute": 0.05
    },
    {
      "callType": "UK Mobile",
      "costPerMinute": 0.08
    },
    {
      "callType": "USA",
      "costPerMinute": 0.10
    },
    {
      "callType": "International",
      "costPerMinute": 0.25
    }
  ]
}
```

Call Types

The destination number determines the call type.

For the purposes of this exercise:

Prefix	Call Type
+44	UK
+1	USA
+447	UK Mobile
+353	Ireland

Anything else International

You may implement this however you choose.

Rating Rules

Duration Charging

Calls should be billed per started minute.

Examples:

Duration	Billable Minutes
1 second	1
60 seconds	1
61 seconds	2
125 seconds	3

Cost Calculation

Example:

125 second UK call

Rate: £0.05 per minute

Billable minutes: 3

Cost: £0.15

Functional Requirements

Store Rate Cards

Ability to create and retrieve customer rate cards.

Rate CDRs

Ability to submit one or more call records for rating.

Produce Rated Output

For each call, return:

- detected call type
- billable minutes
- applied rate
- calculated cost

Example:

```
{  
  "customerId": 1001,  
  "destinationNumber": "+441234567890",  
  "durationSeconds": 125,  
  "callType": "UK",  
  "billableMinutes": 3,  
  "rateApplied": 0.05,  
  "cost": 0.15  
}
```

Suggested API Endpoints

You may structure the API however you prefer, but an example might include:

- create/update custom rate card
- retrieve customer rate card
- submit CDRs for rating

You are free to design the API as you see fit.

Validation Requirements

Please handle invalid inputs appropriately, for example:

- unknown customers
- missing rates
- invalid phone numbers

- negative durations

The API should return sensible HTTP status codes and meaningful error responses.

Testing

Please include automated tests where appropriate.

We are more interested in:

- meaningful test coverage
- maintainability
- clarity

than exhaustive testing.

Documentation

Please include a short README covering:

- how to run the application
 - any assumptions made
 - design decisions
 - trade-offs
 - improvements you would make with more time
-

Submission

Please provide:

- source code
- README

A public Git repository is preferred.

Guidance

This exercise is intentionally open-ended.

We are primarily interested in:

- code quality
- maintainability
- problem solving
- clean architecture
- handling of business rules
- pragmatic engineering decisions

Please avoid unnecessary complexity - a simple, clean, well-structured solution is preferred over overengineering.